

Operations Research

Solving IPs

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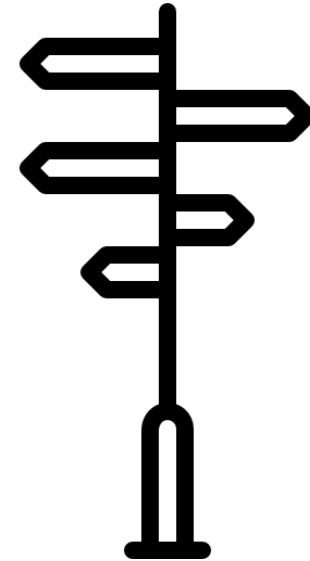
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Course outline

- **Linear Optimization**
 - Modeling by linear programming and graphical solution
 - Solving linear programs, convexity
 - Simplex algorithm, 2-phase simplex
 - Simplex algorithm in matrix notation, goal programming
 - Sensitivity analysis, reoptimization
 - Duality theory, zero-sum games
- **Integer Linear Optimization**
 - Modeling integer programming problems (IPs)
 - **Solving IPs: branch & bound, cutting planes**
 - Advanced solution methods: column generation, approximation algorithms
 - Fast solvable integer problems: total unimodularity, matroids
 - Network flow problems
 - Traveling salesperson problem
- **Nonlinear Optimization**
 - Optimization of multidimensional functions
 - Convex optimization



Today you will learn...

- **solution methods for IPs.**
- how **LPs can be used as bounds** to solve IPs.
- how IPs can be solved by **iteratively removing fractional vertices.**



Enumeration

Explicit enumeration: Compute all possible solutions and choose the best one.

Example:

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 + 4x_3 \\ \text{s. t.} \quad & x_1 + x_2 + x_3 \geq 1 \\ & x_1 + 2x_2 + x_3 \leq 2 \\ & x_1 + x_2 - 2x_3 \geq 0 \\ & x_1, x_2, x_3 \in \{0,1\} \end{aligned}$$

Compute solutions:

x_1	1	1	1	1	0	0	0	0
x_2	1	1	0	0	1	1	0	0
x_3	1	0	1	0	1	0	1	0
feasibility	no	no	no	yes	no	yes	no	no
values				3		2		

Choose the best solution among the possible solutions:

$x_1 = 1, x_2 = 0, x_3 = 0$ with objective value 3

Solving IPs

- The number of possible solutions in the example is $2^3 = 8$.
- If there are n binary variables, then the number of possible solutions is 2^n .

	$n=10$	$n=30$	$n=50$	$n=100$
2^n	10^3	10^9	10^{15}	10^{30}

- This approach works only for a small number of binary variables.
- We want an intelligent enumeration method, which does not list all possible solutions, but only the ones that are meaningful.

“Intelligent enumeration” via Branch-and-Bound

Branching

- Dividing the problem into subproblems
- Subproblems are smaller and easier to solve
- Tree with subproblems as nodes

Bounding

- Constrain the branching process by calculating bounds on the objective function values and exploring subproblems

Logical tests

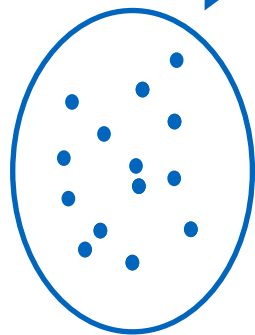
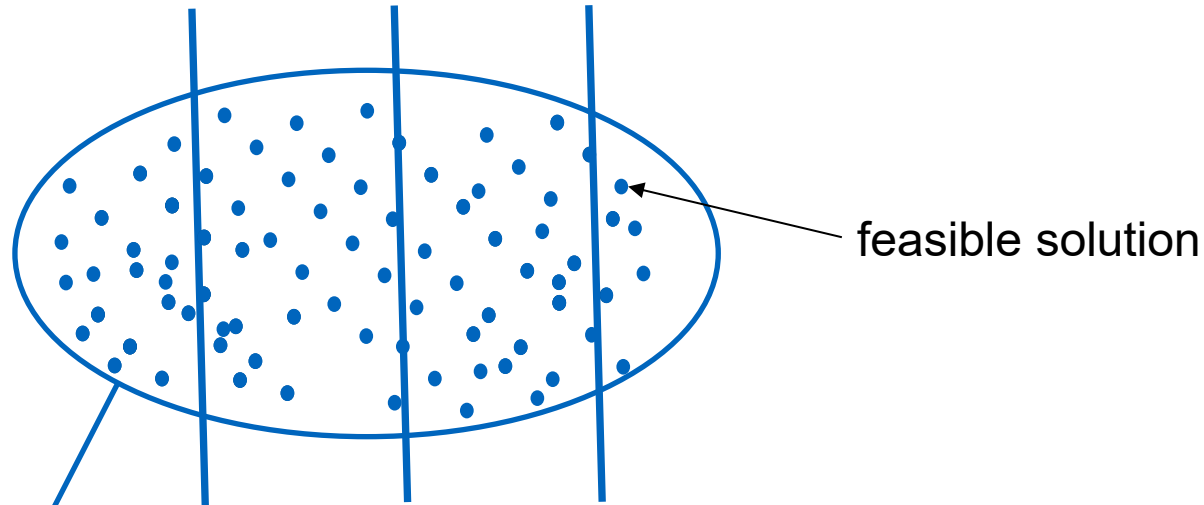
- Modify the problem data and use logical reasoning to further constrain the tree

Solving the subproblems of B&B can be efficiently parallelized across multiple cores or computers!

A. H. Land, A. G. Doig: *An automatic method of solving discrete programming problems*. In: *Econometrica* 28, 1960, S. 497–520.

Branch-and-Bound idea

Solution space:



Smaller part of the solution space:

- Evaluate how good solutions are in this part.
- Intensify search in this part.
- Or search elsewhere if solutions here are certainly worse than in other parts of the solution space.

Branch-and-Bound for general problems

$Z^* :=$ value of a feasible solution ($-\infty$, if none is known)

Divide the feasible space S in disjoint subspaces $S_1, \dots, S_n$
(e.g., by including additional constraints, etc.)

$L = [S_1, \dots, S_n]$

While(L not empty)

- Choose a subspace S' from L .
- Compute an upper bound UB for the optimal value of S' .
- If $UB \leq Z^*$:
 - Remove S' from L
- Else if S' consists of only one feasible element or the solution corresponding to the upper bound is feasible (integral):
 - $Z^* := UB$ and remove S' from L
- Otherwise (if infeasible) divide S' in disjoint subspaces and include them in L .

Relaxing IPs

$$\begin{array}{ll}\max & 8x_1 + 5x_2 \\ \text{s.t.} & x_1 + x_2 \leq 6 \\ & 9x_1 + 5x_2 \leq 45 \\ & x_1, x_2 \geq 0 \text{ and integral}\end{array}$$



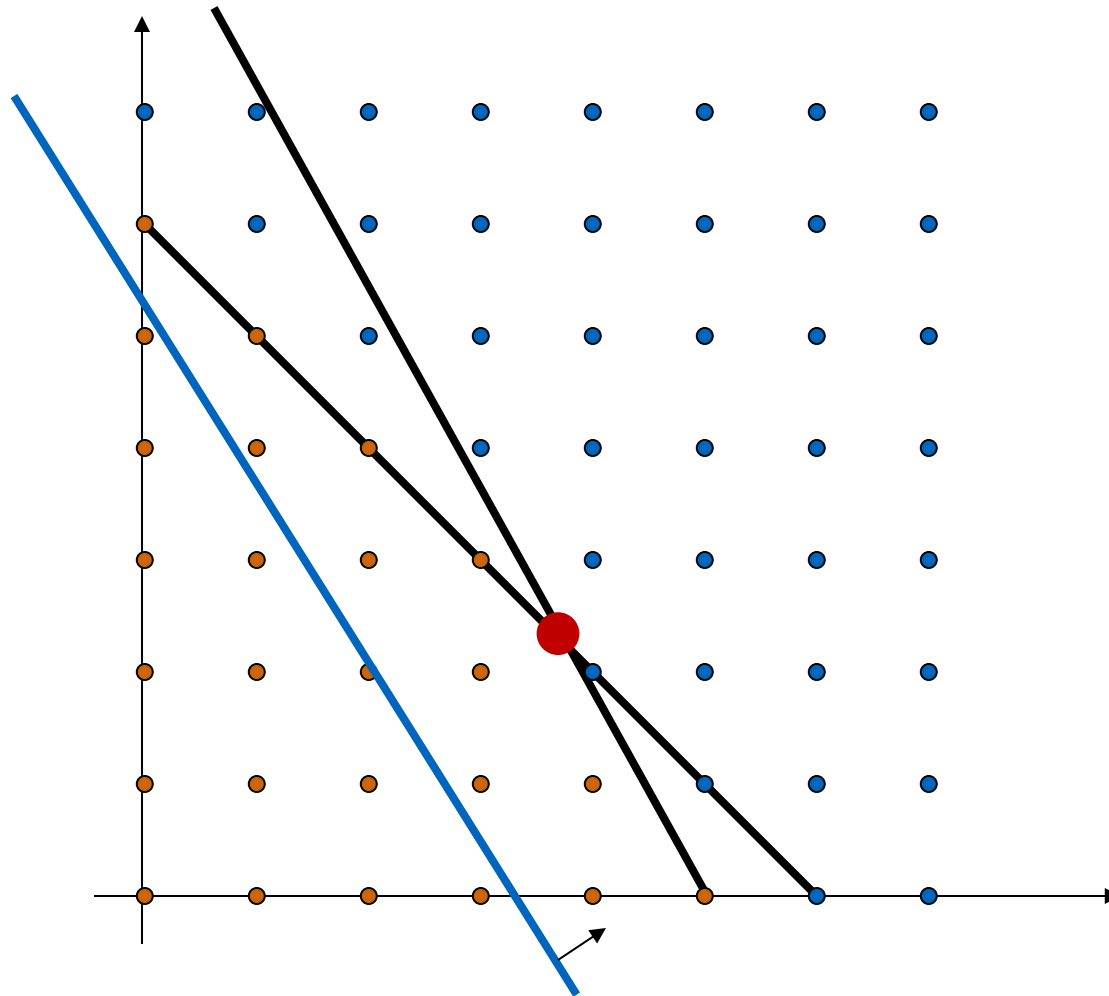
$$\begin{array}{ll}\max & 8x_1 + 5x_2 \\ \text{s.t.} & x_1 + x_2 \leq 6 \\ & 9x_1 + 5x_2 \leq 45 \\ & x_1, x_2 \geq 0\end{array}$$

Optimal solution:

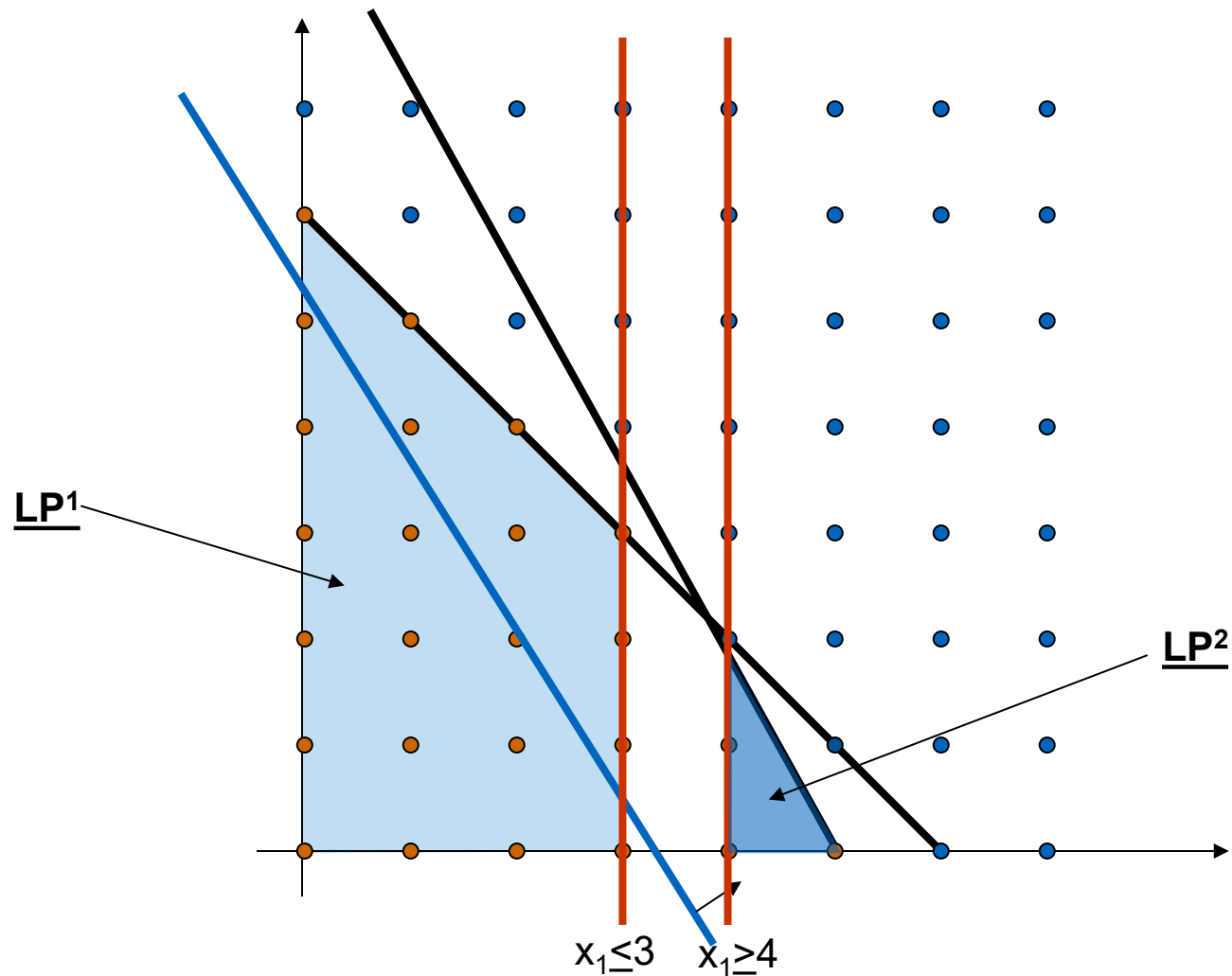
$$x_1 = 3,75, x_2 = 2,25$$

Value: 41,25

Graphical representation



Graphical representation



Branching

Divide the original LP in two LPs:

LP¹: $\max 8x_1 + 5x_2$

$$\text{s.t. } x_1 + x_2 \leq 6$$

$$9x_1 + 5x_2 \leq 45$$

$$x_1 \leq 3$$

$$x_1, x_2 \geq 0$$

Optimal: (3, 3)*

Value: 39

LP²: $\max 8x_1 + 5x_2$

$$\text{s.t. } x_1 + x_2 \leq 6$$

$$9x_1 + 5x_2 \leq 45$$

$$x_1 \geq 4$$

$$x_1, x_2 \geq 0$$

Optimal: (4, 9/5)

Value: 41

Integral! (Further branching is not needed here.)

Best integral solution so far.

Branching at x_2

Divide LP^2 in two parts:

LP³: $\max 8x_1 + 5x_2$

s.t. $x_1 + x_2 \leq 6$

$$9x_1 + 5x_2 \leq 45$$

$$x_1 \geq 4$$

$$x_2 \leq 1$$

$$x_1, x_2 \geq 0$$

Optimal solution: (4,44;1)

Value: $365/9=40,55$

LP⁴: $\max 8x_1 + 5x_2$

s.t. $x_1 + x_2 \leq 6$

$$9x_1 + 5x_2 \leq 45$$

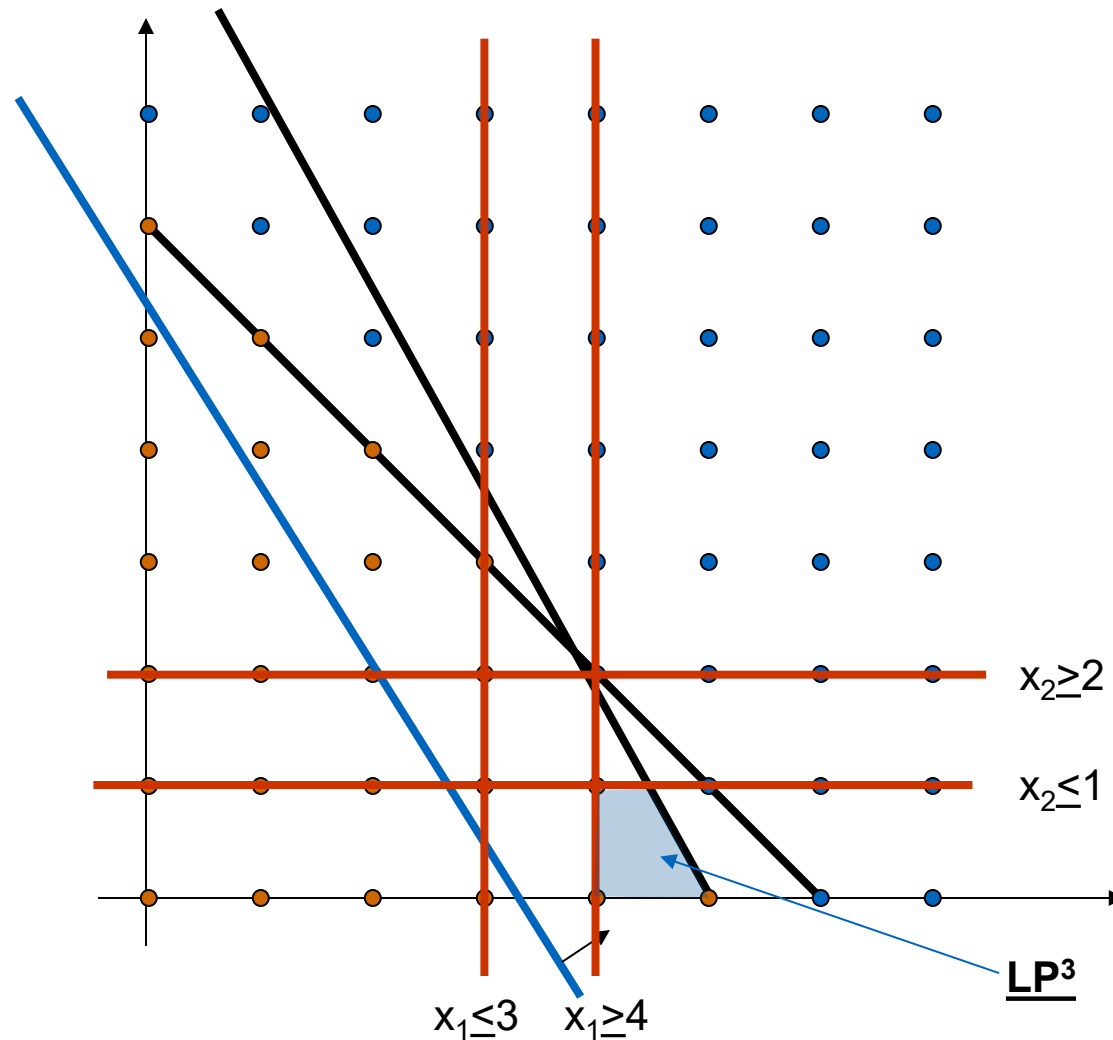
$$x_1 \geq 4$$

$$x_2 \geq 2$$

$$x_1, x_2 \geq 0$$

Infeasible!

Graphical representation



Branching at x_1

Divide LP^3 in:

LP⁵: $\max 8x_1 + 5x_2$

s.t. $x_1 + x_2 \leq 6$

$$9x_1 + 5x_2 \leq 45$$

$$x_1 \geq 4$$

$$x_2 \leq 1$$

$$x_1 \leq 4$$

$$x_1, x_2 \geq 0$$

Solution: $(4, 1)^*$

Value: 37

LP⁶: $\max 8x_1 + 5x_2$

s.t. $x_1 + x_2 \leq 6$

$$9x_1 + 5x_2 \leq 45$$

$$x_1 \geq 4$$

$$x_2 \leq 1$$

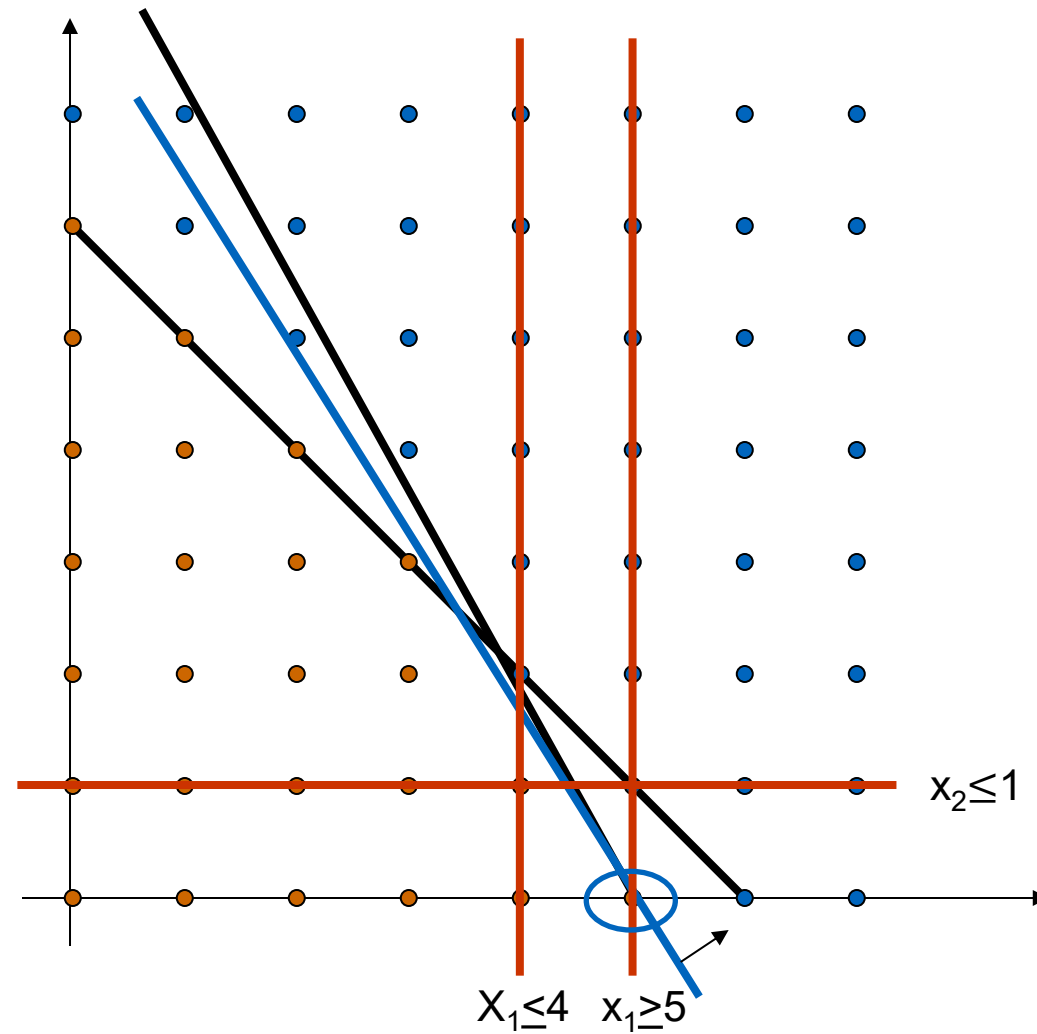
$$x_1 \geq 5$$

$$x_1, x_2 \geq 0$$

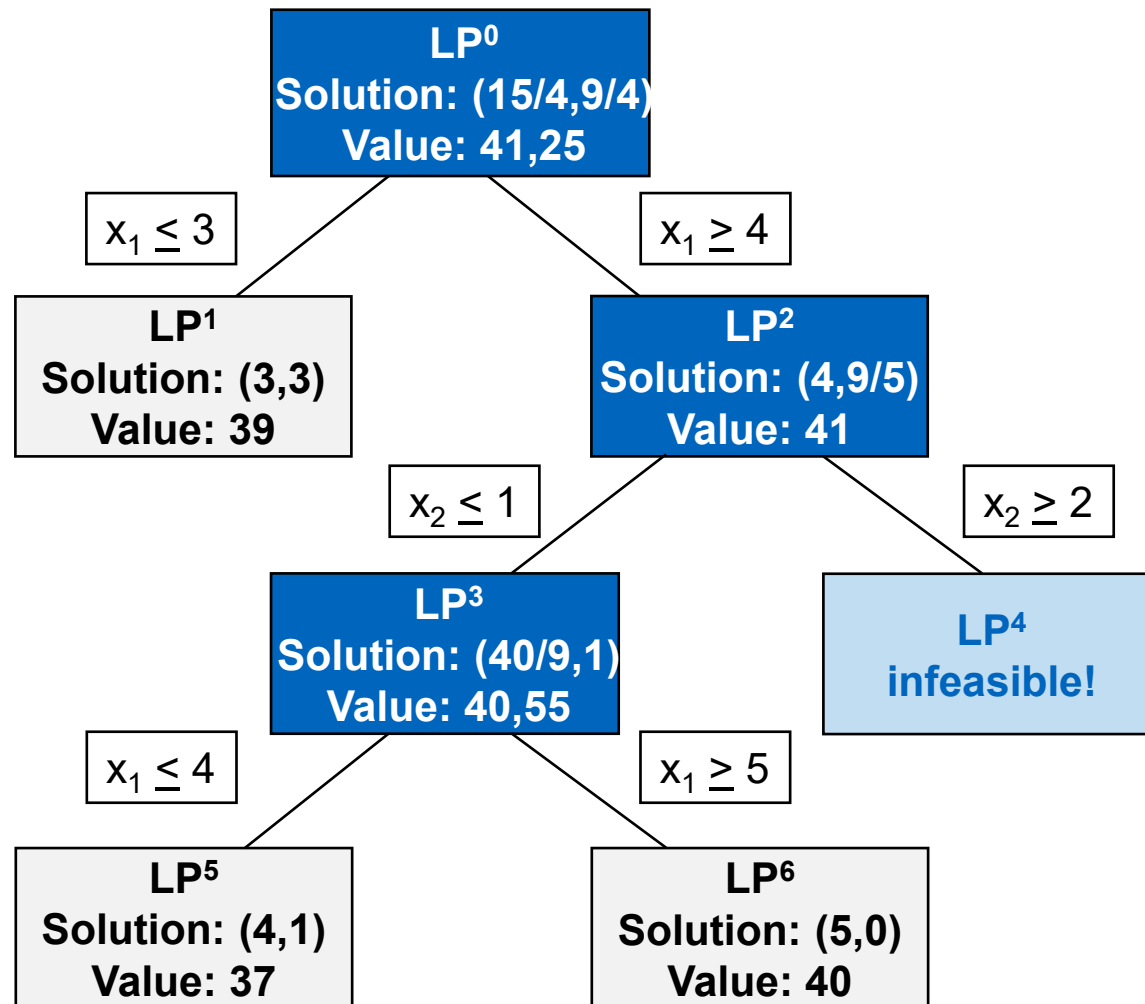
Solution: $(5, 0)^*$

Value: 40

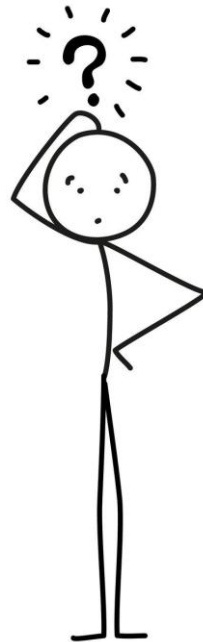
Graphical representation



Branch & Bound tree



Why is it to be expected that the Branch & Bound method finds the optimal solution faster than it is possible by enumerating all solutions?



Philosophy behind (LP-based) B&B

- LPs are used to solve IPs.
- An LP solution provides an upper or lower bound for the IP.
- If the LP solution is integral, the IP is solved.
- If not, the IP is solved by solving two smaller IPs and selecting the better solution (divide & conquer approach).

How to avoid complete enumeration in B&B?

Branching in B&B can be stopped if:

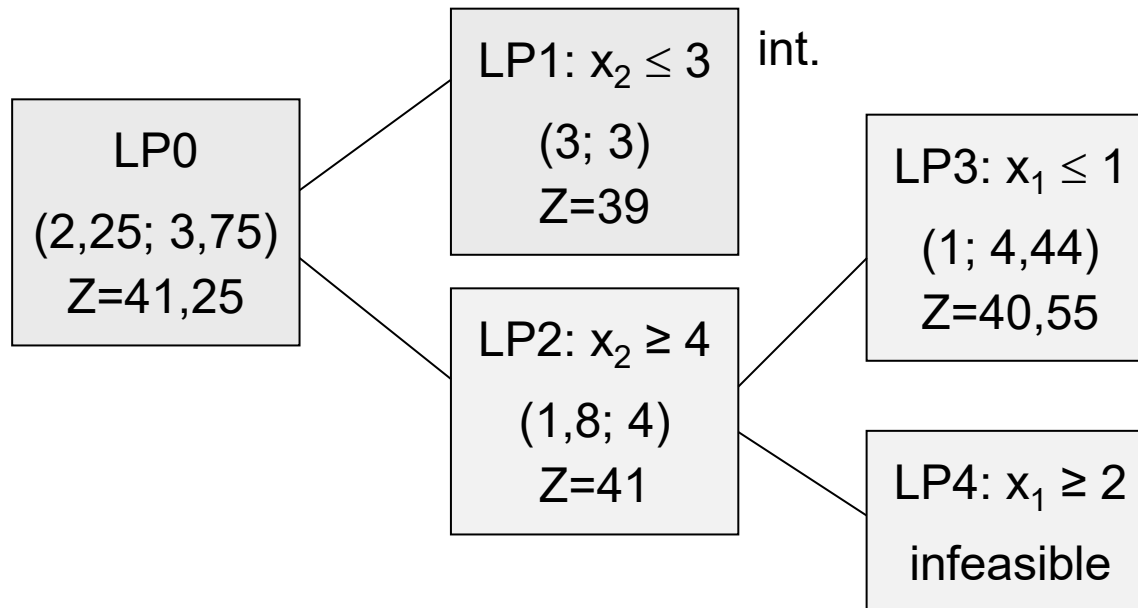
1. The LP relaxation is integral.
2. The LP relaxation is infeasible.
3. The LP relaxation has a lower value than a known IP solution (for maximization problems).

The objective value of the LP relaxation is always at least as large as the objective value of the associated IP problem.

Upper and lower bounds for the best IP solution

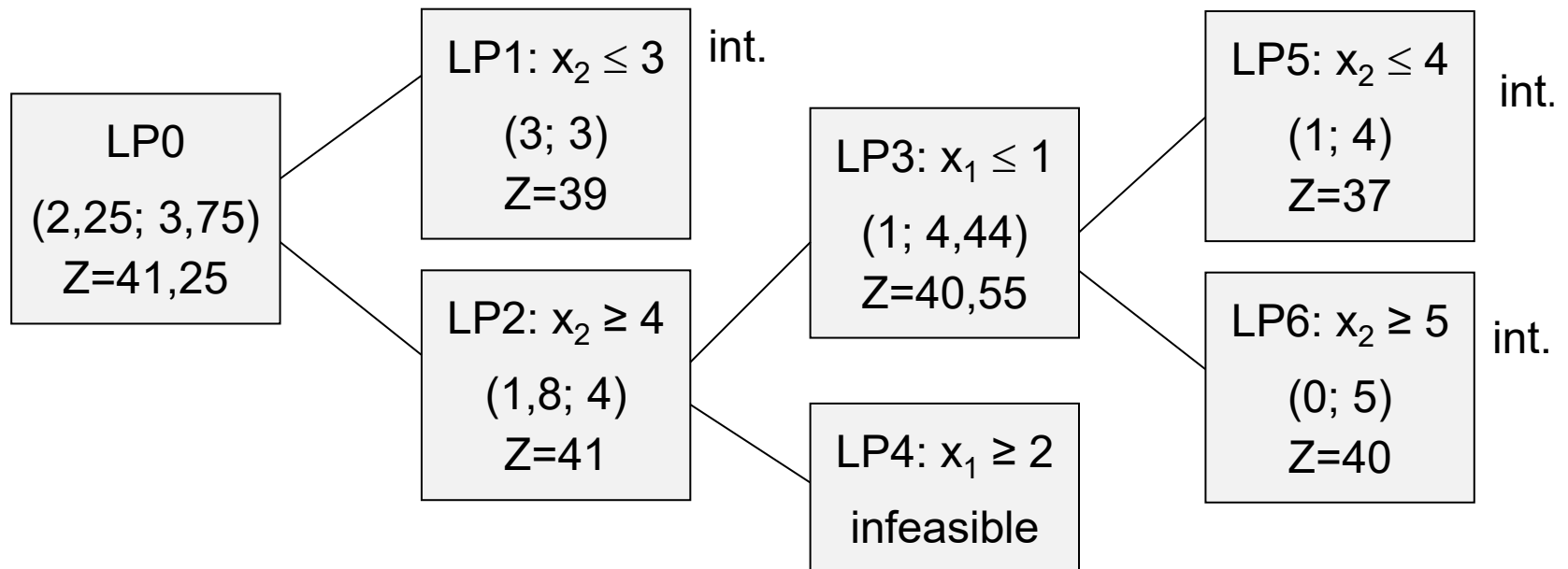
- The current best integral solution (incumbent) is saved. The objective value of the incumbent is denoted by Z^* .
 - Z^* the the best lower bound that we found so far for $\text{OPT}(\text{IP})$: $\text{OPT}(\text{IP}) \geq Z^*$.
- All subproblems are considered, such that
 - the values of the subproblems are larger than Z^* ,
 - and at least one of the child nodes has not been solved yet.
- If Z' is the largest objective value for these subproblems, then Z' is the best upper bound that we found so far: $\text{OPT}(\text{IP}) \leq Z'$.

Example B&B



- If a subproblem is infeasible, its corresponding branch is not investigated further (here subproblem 4).
- Since $39 \leq \text{OPT}(\text{IP}) \leq 40,55$, in subproblem 3 we branch at x_2 .

Example B&B



- If the objective value of a subproblem $\leq Z^*$ (incumbent), the corresponding branch is stopped (subprob. 5).
- If a subproblem is integral and has an objective value $> Z^*$, then x^* becomes the new incumbent (subproblem 6).
- If there are no open subproblems left, then the current incumbent is the optimal solution for IP.

B&B summary

Steps in each iteration:

- 1) **Branching**: Among all open subproblems, choose the one that was considered last. Choose a variable x_i , which had a non-integer value x_i^* in the LP solution of the subproblem. Formulate two new subproblems by adding the constraints $x_i \leq \lfloor x_i^* \rfloor$ and $x_i \geq \lceil x_i^* \rceil$, respectively.
- 2) **Bounding**: Solve the new subproblems and store their LP solutions. Overwrite the current best integral solution (incumbent) if necessary, as well as the lower and upper bounds for the optimal solution $\text{OPT}(\text{IP})$.
- 3) **Test**: For each new subproblem, perform the tests from slide 21. If any of these tests fails, the subproblem is not considered further.
- 4) **Optimality test**: If there are no more open subproblems, the incumbent is returned as the optimum. If there is no such incumbent, the problem is infeasible.
- 5) Otherwise perform another iteration.

Which subproblem should be solved first?

Maximum-Upper-Bound Rule

- The subproblem with the best LP objective value.
- This subproblem is the most promising
 - to obtain a larger IP objective value.
 - to obtain the optimal objective value.

LIFO Rule

- LIFO: Last-In-First-Out, depth-first search
- The last considered subproblem is solved first.
 - This way, we do not have to run the Simplex from the start.
We can reoptimize the super-problem.
- LIFO is often preferred for this reason and quickly leads to a feasible integer solution.

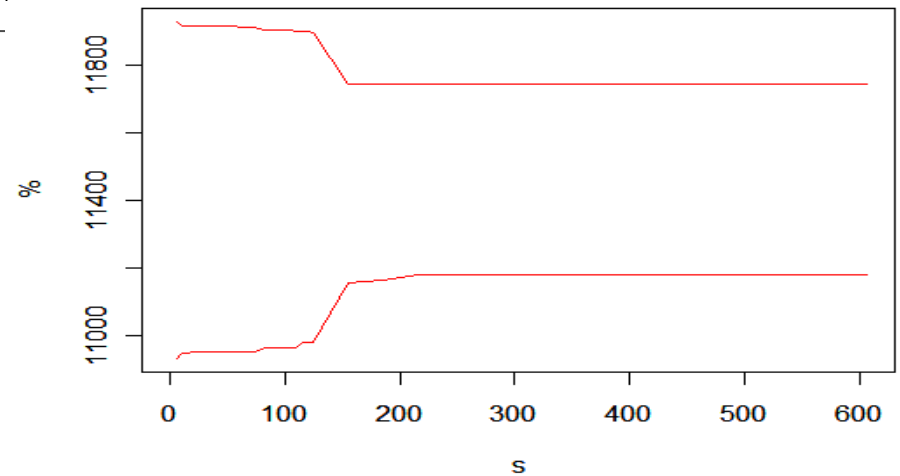
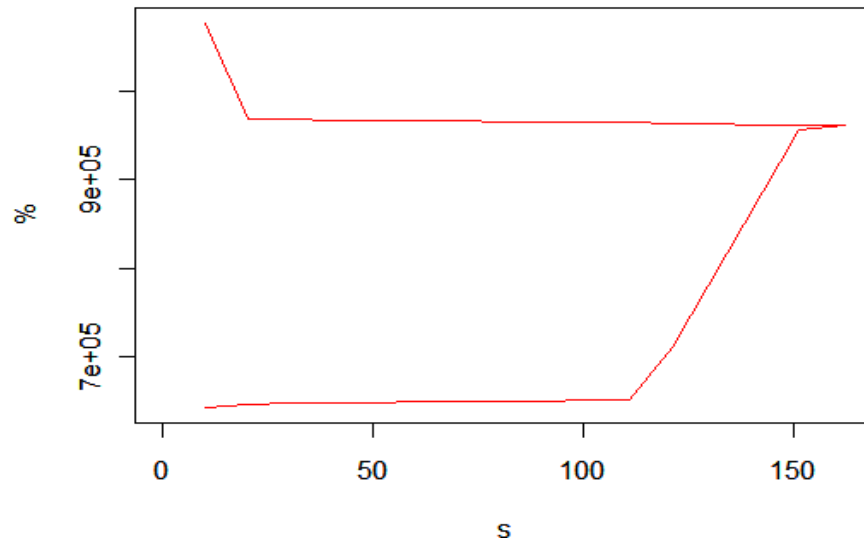
If the solution via B&B takes a long time ...

- B&B can arrive at an optimal solution very quickly but can sometimes arrive at no optimal result after a long time.
- From B&B we know the upper and lower bounds of $OPT(IP)$ in the current iteration: $Z^* \leq OPT(IP) \leq Z'$.
- What can be said about the current best integral solution? If $k = Z' / Z^*$, then:

$$k \cdot Z^* = \frac{Z'}{Z^*} \cdot Z^* = Z' \geq OPT(IP)$$

- This means that the current best integral solution (incumbent) is at most k times worse than the optimal solution. If k is close to 1, then the incumbent is very good.
- For example: if $Z^*=100$, $Z'=102$, then the current incumbent is at most 1,02 times worse than the optimal solution.

Example: lower and upper bounds for minimization



Finding a first integral solution

A current best integral solution (incumbent) is needed to classify the next subproblems.

However, it may take some iterations until the first incumbent is found with B&B.

A first incumbent is therefore often found using a fast heuristic.

Binary programs and MIPs

The only modification for binary programs:

When branching on a binary variable x_i , the subproblems with $x_i = 0$ and $x_i = 1$ are formulated.

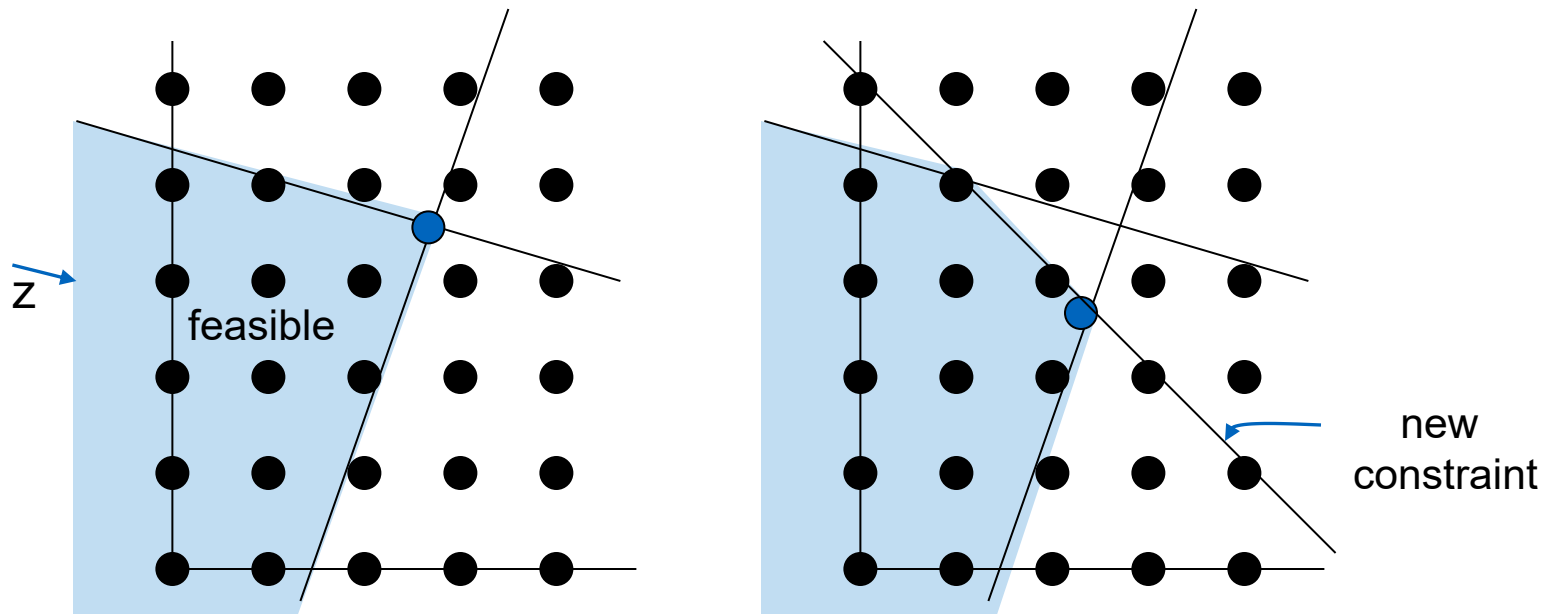
The only modification for MIPs:

Branching is performed only on integer variables.

Summary: B&B for IP

- To solve an IP, solve its LP relaxation.
- If the solution is integral, the IP solution is found.
- If the solution is not integral for the integer variables:
 - Divide the LP into 2 new LPs,
 - solve both LPs recursively,
 - select the best solution.
- Alternative to B&B:
 - Cutting Planes

Cutting planes



- Idea: Cut away the optimal real-valued solution by inserting a new constraint.
- No integral solution is removed in this process.

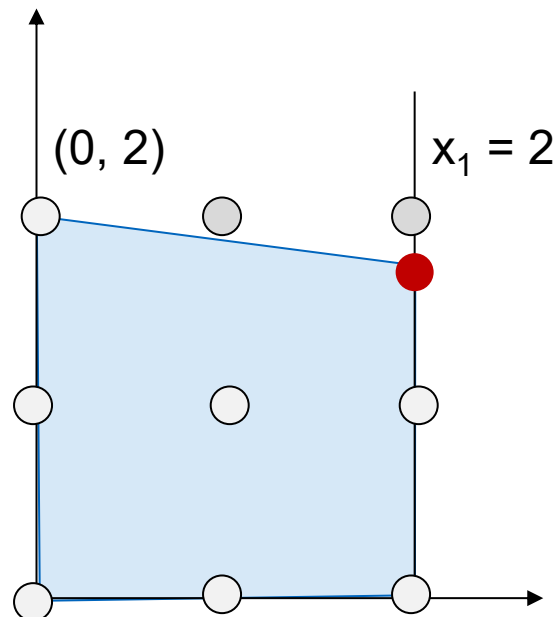
Cutting planes

IP

$$\begin{array}{ll}\max & x_1 + 5x_2 \\ \text{s.t.} & x_1 + 10x_2 \leq 20 \\ & x_1 \leq 2 \\ & x_1, x_2 \geq 0 \text{ integer}\end{array}$$

LP relaxation

$$\begin{array}{ll}\max & x_1 + 5x_2 \\ \text{s.t.} & x_1 + 10x_2 \leq 20 \\ & x_1 \leq 2 \\ & x_1, x_2 \geq 0\end{array}$$



Solution: (2,1.8)
Value: 11

Cutting planes

Include the following constraint: $x_1 + 2x_2 \leq 4$.

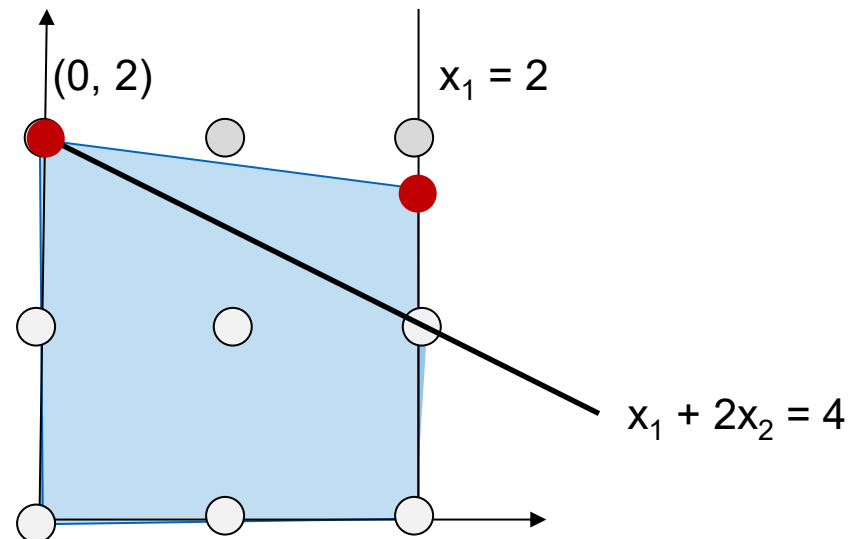
New IP

$$\begin{aligned} \max \quad & x_1 + 5x_2 \\ \text{s.t.} \quad & x_1 + 10x_2 \leq 20 \\ & x_1 \leq 2 \\ & x_1 + 2x_2 \leq 4 \\ & x_1, x_2 \geq 0 \text{ integral} \end{aligned}$$

- The set of feasible points remains the same for the old and the new IP.
- The feasible region of the new LP relaxation has become smaller.
- The optimal solution of the new LP relaxation, $(0,2)$, is integral.

New LP relaxation

$$\begin{aligned} \max \quad & x_1 + 5x_2 \\ \text{s.t.} \quad & x_1 + 10x_2 \leq 20 \\ & x_1 \leq 2 \\ & x_1 + 2x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{aligned}$$



Example for Gomory cuts

New IP

$$\max \quad 8x_1 + 5x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 6$$

$$9x_1 + 5x_2 \leq 45$$

$$x_1, x_2 \geq 0 \text{ integer}$$

Solution of the LP relaxation

$$x_1 = 3,75$$

$$x_2 = 2,25$$

$$Z = 41,25$$

Basic variable	Coefficient					Solution
	Z	x_1	x_2	x_3	x_4	
Z	1			1,25	3/4	41,25
x_2			1	2,25	-1/4	2,25
x_1		1		-1,25	1/4	3,75

Generating cutting cuts

Basic variable	Coefficient					Solution
	Z	x_1	x_2	x_3	x_4	
Z	1			1,25	3/4	41,25
x_2			1	2,25	-1/4	2,25
x_1		1		-1,25	1/4	3,75



$$x_1 - 1,25x_3 + 0,25x_4 = 3,75$$

$$x_1 - 2x_3 + 0,75x_3 + 0x_4 + 0,25x_4 = 3 + 0,75$$

$$x_1 - 2x_3 + 0x_4 - 3 = 0,75 - 0,75x_3 - 0,25x_4$$

Choose a constraint in which a basic variable takes a non-integer value.

Eliminate fractional parts on the left side and create right side with negative fractional parts of the variables and positive fractional part of the b' vector.

Generating cutting cuts

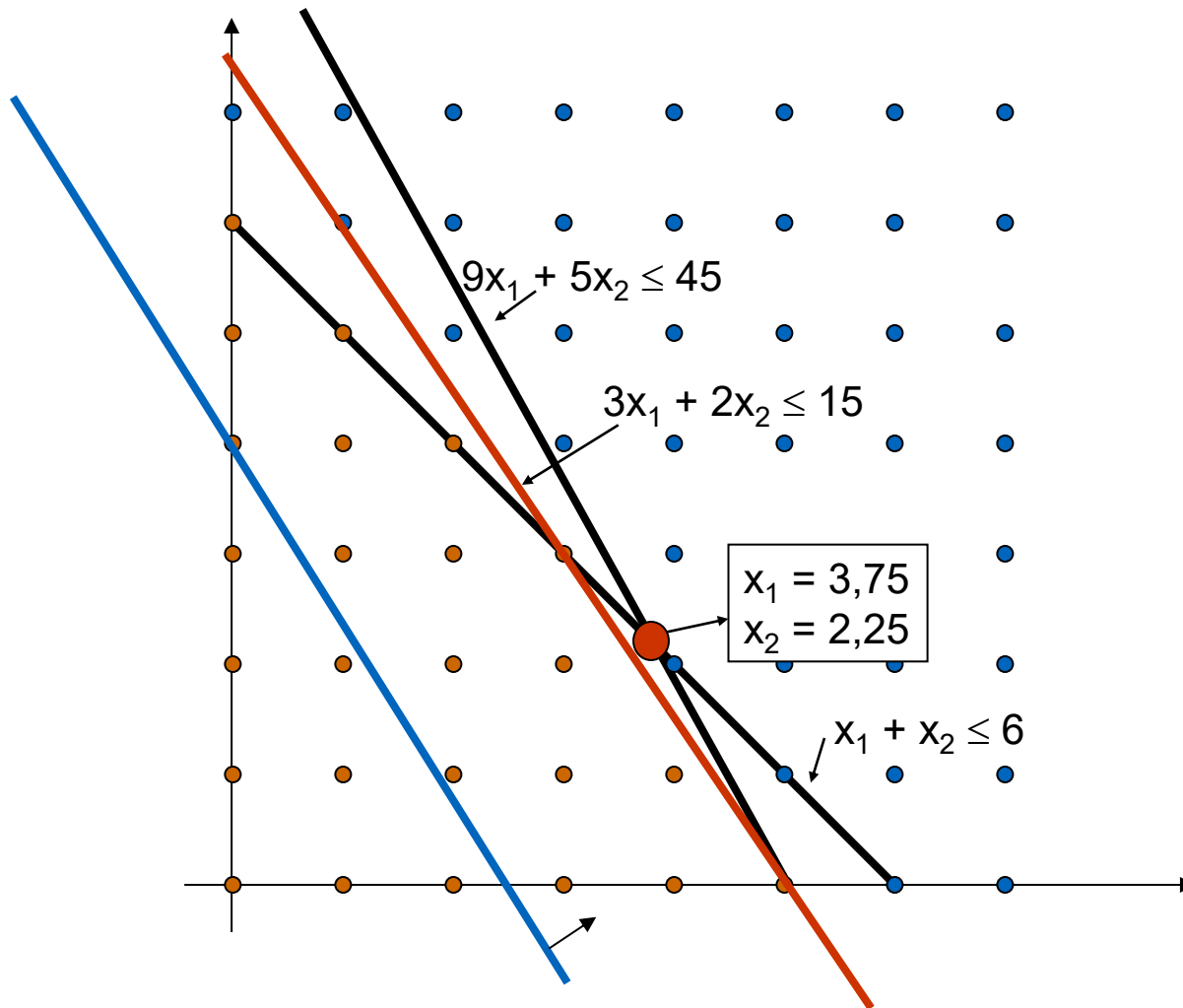
$$x_1 - 2x_3 + 0x_4 - 3 = 0,75 - 0,75x_3 - 0,25x_4$$

- For any integral point, the left-hand side must also be integer.
- In any feasible solution, the slack variables x_3 and x_4 must be ≥ 0 and integral.
- Any feasible solution to the right side must be less than 1, since $0,75 < 1$.
- Thus, the right side must be integral and less than 1.
- This results in a cutting plane that does not remove any integer solution:

$$0,75 - 0,75x_3 - 0,25x_4 \leq 0$$

- The current optimal solution for the LP relaxation has $x_3 = x_4 = 0$ and cannot fulfill the above cutting plane ($0,75 > 0$), i.e. a part of the polytope is cut off.

Graphical representation



$$0,75 - 0,75x_3 - 0,25x_4 \leq 0$$

Converting the
slack variables:

$$x_3 = 6 - x_1 - x_2$$

$$x_4 = 45 - 9x_1 - 5x_2$$

$$\rightarrow 3x_1 + 2x_2 \leq 15$$

Including the cutting plane

... $0,75 - 0,75x_3 - 0,25x_4 \leq 0$ to the last optimal tableau with the new slack variable x_5

Basic variable	Coefficient						Solution
	Z	x_1	x_2	x_3	x_4	x_5	
Z	1			1,25	3/4		41,25
x_2			1	2,25	-1/4		2,25
x_1		1		-1,25	1/4		3,75
x_5				-0,75	-1/4	1	-0,75

Solution

Basic variable	Coefficient						Solution
	Z	x_1	x_2	x_3	x_4	x_5	
Z	1				0,33	1,67	40
x_2			1		-1	3	0
x_1		1			-0,16	-1,67	5
x_3				1	0,33	-1,33	1

Cutting planes method

- Compute the LP relaxation.
- Choose a constraint from the optimal Simplex tableau whose non-integer part of the right-hand side is close to 0.5.
- Reformulate the constraint from the optimal tableau:
All terms with integer coefficients
= all terms with non-integer coefficients
- Derived cutting plane:
 - all terms with non-integer coefficients ≤ 0
- Solve the extended LP relaxation via the (dual) simplex.
- Above steps are repeated until the solution is integral.

A short comment on the Dual Simplex method

Start with a basis that might not be a feasible primal solution.

$$\begin{array}{ll}\min & 3x_1 + 2x_2 \\ \text{s. t.} & x_1 + x_2 \geq 1 \\ & 2x_1 - 3x_2 \geq 2 \\ & x_1, x_2 \geq 0\end{array}$$

Standard form:

$$\begin{array}{ll}\max & -3x_1 - 2x_2 \\ \text{s. t.} & x_1 + x_2 - x_3 = 1 \\ & 2x_1 - 3x_2 - x_4 = 2 \\ & x_1, \dots, x_4 \geq 0\end{array}$$

Example

With basis $\{x_3, x_4\}$ the tableau is optimal, but not feasible (neg. solution).

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1	3	2			0
x_3		-1	-1	1		-1
x_4		-2	3		1	-2

Exiting basic variable (with largest neg. value): x_4

Entering basic variable: the variable with negative coefficient

If multiple are possible, choose the variable with the largest coefficient
(row 1/pivot row: $3/-2$)

Example

- Final tableau after one iteration: all basic variables are positive.

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1		$13/2$		$3/2$	-3
x_3			$-5/2$	1	$-1/2$	0
x_1		1	$-3/2$		$-1/2$	1

- If there were a constraint in which the result column is negative and every variable in the row has a non-negative coefficient, then the LP is infeasible.

Branch-and-Cut algorithm

- Integer programs are rarely solved using the cutting plane method.
- Often B&B and the cutting plane methods are combined to solve IPs.
- Cutting planes are added to subproblems of B&B to make the process faster. When this is no longer possible, branching is used.
- This combination is also called the branch-and-cut algorithm.

Solution approaches

Common approaches for finding optimal solutions:

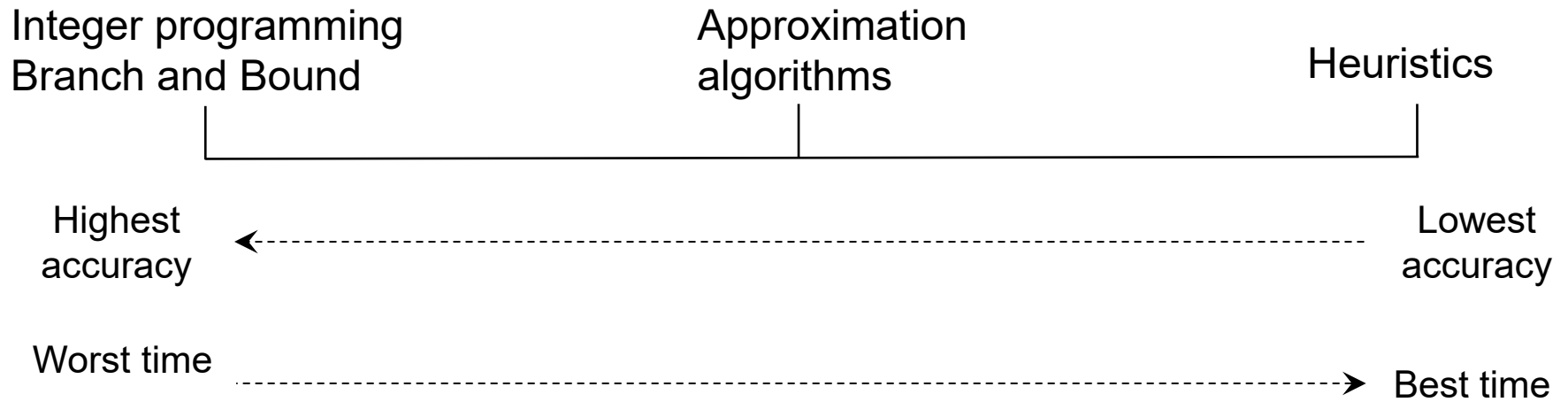
- Cutting Planes
 - Gomory, 1958; etc.
- Branch-and-Bound
 - Land & Doig, 1960
- Branch-and-Cut
 - Manfred Padberg, Martin Grötschel, 80s, 90s until today
- Dynamic optimization

Alternative for large IPs:

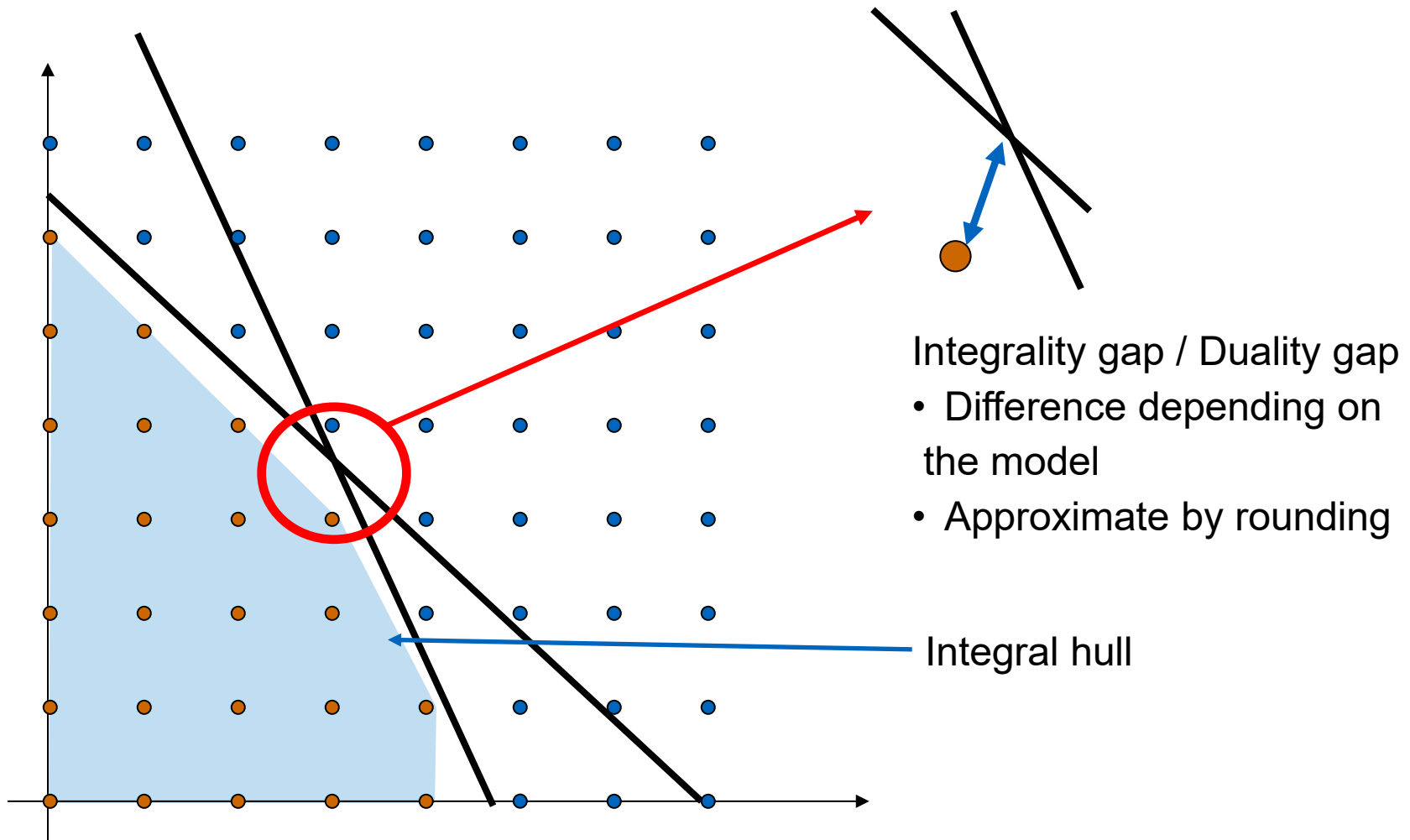
- Approximation algorithms
- Heuristics and metaheuristics
 - usually do not compute optimal solutions, but can be fast
 - considering problem specific information is important (no free lunch theorem)

Solving NP-hard discrete optimization problems

- Integer programming
- Approximation algorithms
- Heuristics



Integrality gap



IP duality

z = optimal objective value of the IP

z' = optimal objective value of the relaxed IP (LP)

q' = optimal objective value of the dual P'

$$z \leq z' = q'$$

Duality gap = $z' - z$

- In IPs, constraints with slack also have economic value.
- This means that the concept of marginality is not easily applicable to discrete choices.